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Jiandun Li, Huiyao Zheng, Yabin Zhou & Fusong Jiang

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A bimodal dataset for diabetes research

Jiandun Li[1], Huiyao Zheng[1], Yabin Zhou[1], Fusong Jiang[2, 3, 4*]

[1] School of Electronic Information Engineering, Shanghai Dianji University, Shanghai 201306, China

[2] Department of Endocrinology and Metabolism, Shanghai Jiao Tong University School of Medicine Affiliated Sixth People's Hospital, Shanghai 200233, China

[3] Shanghai Diabetes Institute, Shanghai 200233, China

[4] Shanghai Clinical Center for Diabetes, Shanghai 200233, China

Abstract: In recent years, with the continuous booming of diabetes patients, the research on diabetes and its complications, including pathogenesis, early diagnosis and therapeutic interventions, has attracted considerable attention. However, the lack of large-scale real datasets has significantly impeded its in-depth development. To address this challenge, we hereby disclose our diabetes dataset of 5,922 examples and 190 attributes, spanning across many detailed and well-curated clinical and demographic records, e.g., BMI, lifestyle factors, family history, glycemic control, insulin-related measures, lipid and metabolic profiles, which can shed light for reliable analyses of diabetes progression, complication risks and associated physiological and metabolic factors.

Background & Summary

As a severe global crisis for public health, diabetes manifests pathological heterogeneity through distinct categories, i.e., type 1 diabetes (T1DM), type 2 diabetes (T2DM) and gestational diabetes, and systemic metabolic dysregulation could trigger end-organ complications such as retinopathy, diabetic nephropathy and cardiovascular emergency [1, 2, 3]. Therefore, early diagnosis of diabetes together with its complications is of great importance to manage the lifecycle of this illness. To address this issue, researchers have introduced many a machine learning model.

For instance, based on multiple health attributes, Febrian et al [4] proposed a comparative analysis via K-Nearest Neighbors (KNN) and Naïve Bayes for diabetes prediction. Through 618 participants' records, Fagherazzi et al [5] applied the Discriminative Dimensionality Reduction with Trees and revealed the heterogeneity of glycemic phenotypes in T1DM, and identified seven phenotypes and their related determinants. Based on a diabetic retinopathy dataset, Gadekallu et al [6] proposed a Deep Neural Network model optimized by Principal Component Analysis and Grey Wolf Optimization. With the cardiovascular complications in T2DM centered, Kee et al [7] conducted a systematic review on related machine learning models, including Support Vector Machine (SVM), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Gaussian Naïve Bayes (Gaussian NB), KNN, deep neural network, extreme gradient boosting.

To leverage the capability of machine learning, a great quantity of clinic records from real hospitals with rich attributes and annotations are necessary. Recently, Prioleau et al [8] provided a DiaTrend dataset, which consists of intensive longitudinal data from wearable medical devices, including 27,561 days of continuous glucose monitoring data and 8,220 days of insulin pump data from 54 patients with diabetes. However, the dataset does not have any attributes related to diabetes complications. Additionally, Salcedo-Tacuma et al [9] collected multiple data samples on transcriptomic features of subcutaneous and omental adipose tissues from

* Corresponding author: Fusong Jiang, E-mail: hajfs@126.com.

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patients with gestational diabetes and matched controls during cesarean delivery using next generation RNA sequencing. Nevertheless, it only took 5 patients into consideration, which is insufficient to generalize to real-world cases. Rodriguez-Leon et al [10] provided a dataset that encompass continuous glucose level measurements, alongside patient demographic and clinical data. These examples are sourced from 736 T1DM patients in Spain, and the time spans across 257,780 days over a four-year period. Although this dataset is multimodal, important attributes, such as diabetes history are still missing. Facing the challenge of limited 3D-spatial and quantitative information, Hahn et al [11] present a dataset consisting of optical images of full β-cell subject to diverse doses of administration. More digital features such individual islet β-cell volumes, spatial coordinates, and shape together with signal intensities for insulin are provided. Nevertheless, this dataset is comprehensive, significant attributes are missing, e.g., single-cell transcriptome and proteome data, therefore it has little impact on diabetes diagnosis and complication detection on humans. To visually summarize their comparative advantages and existing gaps, a detailed overview is provided in Table 1.

Datasets	**Data Scale**	**Data Modality**	**Key Strengths**	**Major Limitations**
Prioleau et al [8]	54 patients 27,561 days of CGM data	Continuous Glucose Monitoring (CGM) Insulin Pump Data	Large-scale, longitudinal physiological monitoring	Lacks attributes related to diabetes complications
Salcedo-Tacuma et al [9]	5 patients	Transcriptomic Data (Adipose Tissue)	Provides molecular-level insights from specific tissues	Extremely small sample size limits generalizability
Rodriguez-Leon et al [10]	736 T1DM patients 257,780 days of data	Demographic & Clinical Data Continuous Glucose Measurements	Large-scale, multimodal, and long-term data collection	Missing critical clinical attributes (e.g., diabetes history)
Hahn et al [11]	β -cell images	Optical Imaging Cellular Morphology & Spatial Data	Provides 3D spatial and quantitative cellular information	Lacks clinically translatable data (e.g., transcriptomics); Not applicable for human diagnosis
bimodal dataset [12] (This Work)	5,922 instances 190 attributes	Multidimensional Clinical Parameters	Comprehensive attributes (e.g., BMI, Fasting Blood Glucose) ; Includes detailed diabetes history; Contains data on diabetes complications	-

Table 1. Comparative Analysis of Publicly Available Diabetes-Related Datasets

Although these datasets have successfully supervised some machine learning models for diabetes related researches, there still exist several limitations: (1) the number of patients included is limited (averaged 265); (2) the feature dimensions are also inadequate (averaged 17.5); (3) they are always monomodal-oriented, i.e., in the format of real number. As a response, this study systematically builds a bimodal (digital and text) diabetes dataset

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[12] supported by real-world diabetes patients, including 190 attributes and 5,922 samples recorded in two months. It can facilitate related researches on diabetes pre-diagnosis, complication prediction and treatment adherence. The characteristics of this dataset is threefold: (1) it captures multi-system metabolic indicators, e.g., liver and kidney functions; (2) a wide range of patients, i.e., aging from 18 to 91 years old, are covered; (3) multiple complications, such as diabetic nephropathy and blindness are involved.

Methods

Participants. The data in this study were derived from the de-identified cross-sectional dataset of diabetes patients in the Department of Diabetes and Endocrinology at Shanghai Jiao Tong University School of Medicine Affiliated Sixth People's Hospital from February to April, 2012. The dataset [12] encompasses demographic and clinical characteristics of diabetic patients, including fundamental traits and obesity indices, with a focus on their correlations and mechanistic impacts on diabetes onset. Detailed demographic information and basic physiological profiles of diabetic patients are also provided. In general, this dataset [12] can enable in-depth analysis of the specific effects of glycemic status, lipid profiles, insulin levels, hepatic, renal and hematologic parameters, integrated with disease history, lifestyle factors, complication profiles and genetic predispositions, along the development and progression of diabetes mellitus.

Data Cleaning and Preparation. To prepare the dataset, the following steps were performed: (1) renaming variables: original variables were uniformly converted into standardized English nomenclature. (2) de-identification: all personal identifiers have been removed to secure privacy.

Ethics Approval. This study was approved by the Ethics Committee of Shanghai Jiao Tong University School of Medicine Affiliated Sixth People's Hospital (Approval No. 2019-022).

Consent for Data Sharing. The Committee specifically reviewed and granted a waiver of informed consent for this retrospective analysis of fully de-identified medical records. This determination is documented in the associated Chair Reviewer's Assessment Form for Waiver of Informed Consent. Given the comprehensive de-identification of the data and the minimal risk involved, the sharing of this anonymized dataset for research purposes is within the scope of this ethical approval.

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Data Records

All data in this dataset [12] are stored and accessible via the Zenodo platform, comprising a single Excel file with 190 variables and 5,922 anonymized patient records. A detailed data dictionary explaining all variables is provided in the Supplementary Information. This multidimensional resource encompasses demographic characteristics, physical examinations, biochemical measurements, medical histories, lifestyle factors, and complication profiles. Collected during February to April 2012, the dataset supports epidemiological investigations and clinical analyses of diabetes, metabolic syndrome, and related comorbidities.

Blood test indicators. Table 2 presents indicators related to islet function, covering 4 key attributes, i.e., FINS, Fasting C-peptide (FCP), 2-Hour Insulin (INS2h), and 2-Hour C-peptide (CP2h).

Biochemical Parameter	Description
FINS	Fasting Insulin
FCP	Fasting C-peptide
INS2h	2-Hour Insulin
CP2h	2-Hour C-peptide

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Table 2. Table of Pancreatic Islet Function Indicators

Diabetes judgment indicators. Table 3 illustrates 4 diabetes judgment indicators, including the Diabetes Mellitus (DM), the New Diabetes Diagnosis Method (DMnew3class), the Three-category Classification of Blood Glucose (PGclass3) and the Six-category Classification of Blood Glucose (PGclass6).

Biochemical Parameter	Description
DM	Diabetes: 0 = 'no', 1 = 'yes'
DMnew3class	Characteristics of newly diagnosed diabetes: 1 = 'Diabetes diagnosed via fasting blood glucose', 2 = 'Diabetes diagnosed via 2-hour blood glucose', 3 = 'Diabetes diagnosed via both fasting and 2-hour blood glucose'
PGclass3	Three-category classification of blood glucose: 0 = 'NGR (Normal Glucose Regulation)', 1 = 'Prediabetes', 2 = 'Diabetes Mellitus (DM)'
PGclass6	Six-category classification of blood glucose: 0 = 'NGR (Normal Glucose Regulation)', 1 = 'IFG (Impaired Fasting Glucose)', 2 = 'IGT (Impaired Glucose Tolerance)', 3 = 'IFG＆IGT (Combined Impaired Fasting Glucose and Impaired Glucose Tolerance)', 4 = 'NDM (Neonatal Diabetes Mellitus)', 5 = 'KDM (Ketosis-prone Diabetes Mellitus)'

Table 3. Table of Diabetes Classification Criteria

Liver function. Table 4 consists of 15 attributes, such as Alanine Aminotransferase (ALT), ALT Segmentation (ALTn), Aspartate Aminotransferase (AST), etc. These are key markers for gauging the liver's metabolic, detoxification, and synthetic functions.

Biochemical Parameter	Description
ALT	Alanine Aminotransferase
ALTn	ALT Segmentation: 0 = "<40", 1 = "≥40"
AST	Aspartate Aminotransferase
ASTALT	AST/ALT
GGT	Gamma-Glutamyl Transferase（γGT）
ALP	Alkaline Phosphatase
TBILn	Total Bilirubin Elevation Segmentation: 0 = "≤17. 1", 1 = ">17. 1"
TBIL	Total Bilirubin
DBIL	Direct Bilirubin
IDBIL	Indirect Bilirubin
TP	Total Protein
ALB	Albumin
GLO	Globulin
AG	A/G

Table 4. Liver Function Indicators

Kidney function. Table 5 encompasses 22 attributes, including Blood Urea Nitrogen (BUN), Serum

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Creatinine (SCRE), and the Renal Insufficiency (highCr).

Biochemical Parameter	Description
BUN	Blood Urea Nitrogen (BUN)
SCRE	Serum Creatinine
highCr	Renal Insufficiency: 0 = "None", 1 = "Serum Creatinine > 106"
SUA	Serum Uric Acid
highUA	Serum Uric Acid Elevation: 0 = "Normal", 1 = "Elevated"
UCRE	Urine Creatinine
UmALB	Urine Microalbumin
highUmALB	Urine ALB Elevation: 0 = "Normal", 1 = "Elevated"
UMAUCR	Urine MA/UCR (Urine Microalbumin-to-Creatinine Ratio)
ACR3degree	ACR (Albumin-to-Creatinine Ratio) Three-Tier Classification: 1 = "<30", 2 = "≥30 & <300", 3 = "≥300"
highACR	ACR Elevation: 0 = "<30", 1 = "≥30"
GFR	Calculated GFR (Glomerular Filtration Rate)
GFR5lev	GFR Five-Tier Classification (GFR5lev): 0 = "≥90", 1 = "<90", 2 = "<60", 3 = "<30", 4 = "<15"
GFR6levG5	GFR3a3b: 1 = "eGFR≥90 mL/min/1.73m^2", 2 = "eGFR<90 mL/min/1.73m^2 &≥60 mL/min/1.73m^2 ", 4 = "eGFR<30 mL/min/1.73m^2 &≥15 mL/min/1.73m^2 ", 5 = "eGFR<15 mL/min/1.73m^2 ", 31 = "eGFR<60 mL/min/1.73m^2 &≥45 mL/min/1.73m^2 ", 32 = "eGFR<45 mL/min/1.73m^2 &≥30 mL/min/1.73m^2 "
EUGFRabACR012	eGFR normal urine ACR segmentation: 0 = "eGFR ≥ 90 mL/min/1.73m^2 and ACR < 30", 1 = "eGFR ≥ 90 mL/min/1.73m^2 and ACR ≥ 30", 2 = "eGFR < 90 mL/min/1.73m^2"
EUGFR90UACR01	eGFR less than 60 mL/min/1.73m^2 or UACR greater than or equal to 30mg/g： 0 = "eGFR ≥ 60 mL/min/1.73m^2 and UACR < 30 mg/g", 1 = "eGFR < 60 mL/min/1.73m^2 or UACR ≥ 30mg/g"
EUGFR60abACR012	eGFR ≥ 60 normal urine ACR segmentation: 0 = "eGFR ≥ 60 mL/min/1.73m^2 and ACR < 30", 1 = "eGFR ≥ 60 mL/min/1.73m^2 and ACR ≥ 30", 2 = "eGFR < 60 mL/min/1.73m^2 "
UACRGFR	Positivity defined as urine ACR ≥ 30 and eGFR < 60 mL/min/1.73m^2 : 0 = "Negative", 1 = "Positive"
DN	Diabetic nephropathy staging: 0 = "Urine < 30 mg"， 3 = "Stage 3 ≥ 30 mg", 4 = "Stage 4 ≥ 300 mg with GFR < 10 mL/min/1.73m^2 ", 5 = "Stage 5 uremia"

Table 5. Renal Function Indicators

Blood routine. Table 6 encompasses 27 attributes, including White Blood Cells (WBC), Red Blood Cells

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(RBC), Hemoglobin (HGB), etc.

Biochemical Parameter	Description
WBC	White Blood Cells (WBC)
LC	Percentage of Immature Cells
LCcount	Absolute Count of Immature Cells
ALY	Percentage of Atypical Lymphocytes
ALYcount	Absolute Count of Atypical Lymphocytes
RBC	Red Blood Cells (RBC)
RDWCV	RDW-CV
RDWSD	RDW-SD
MCV	Mean Corpuscular Volume (MCV)
HCT	Hematocrit (Hct)
LYMPH	LYMPH%
LYMPHcount	Lymphocyte Count
MCH	Mean Corpuscular Hemoglobin (MCH)
MCHC	Mean Corpuscular Hemoglobin Concentration (MCHC)
MPV	Platelet Mean Volume (MPV)
BASO	BASO%
BASOcount	Basophil Count
EO	EO%
EOcount	Eosinophil Count
HGB	Hemoglobin (Hb)
Hb4per	Hemoglobin 4 Classification
PLT	Platelet (PLT)
PDW	PDW
PCT	PCT
MONO	MONO%
MONOcount	Monocyte Count
NEUT	NEUT%
NEUTcount	Neutrophil Count

Table 6. Blood Routine Indicators

Medical history and lifestyle. Table 7 gives 23 attributes, documenting crucial medical history information, such as a History of Diabetes (DMhis) and a History of Hypertension (hypertenhis); meanwhile, it conducts multidimensional quantitative assessments of smoking and alcohol, e.g., frequency, starting age and duration.

Biochemical Parameter	Description
DMhis	History of Diabetes: 0 = 'no', 1 = 'yes'
CDSMS	CDS Metabolic Syndrome: 0 = 'no', 1 = 'yes'
Hypertenhis	History of Hypertension: 0 = 'no', 1 = 'yes'
Dyslipidehis	History of Dyslipidemia: 0 = 'no', 1 = 'yes'

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smoking2	Categorized as 'no' for occasional smoking: 0 = 'no', 1 = 'yes'
smoking	Smoking: 1 = 'no', 2 = 'occasionally', 3 = 'regularly'
smokingAge	Age started smoking
smokingRange	Years of smoking
drking2	Categorized as 'no' for occasional drinking: 0 = 'no', 1 = 'yes'
drking	Alcohol consumption: 1 = 'none', 2 = 'occasionally', 3 = 'regularly'
drkingAge	Age started drinking
drinkg	Amount per occasion (liang)
escdrink	Ever quit drinking
escdmon	How many months since quitting
escdti	Number of times quit
escdbe	Reason for quitting
escdbeseek	Name of the disease leading to quitting
escdoter	Other reasons for quitting
drinkeyears	Total years of drinking
tea	Tea consumption: 1 = 'no', 2 = 'occasionally', 3 = 'regularly'
worktype	Type of work: 1 = 'predominantly physical', 2 = 'equal physical and mental', 3 = 'predominantly mental', 9 = 'unknown'

Table 7. Medical History and Lifestyle

Complications. Table 8 shows 22 attributes, e.g., Carotid Artery Disease (CAP), Hyperuricemia (hyperUA), Cerebral Infarction (Cerebralin), Intracerebral Hemorrhage (Cerebralhe). Additionally, the diabetes statuses of relatives including fathers, mothers, siblings and children are also included.

Biochemical Parameter	**Description**
angina	History of Angina: 0 = 'no', 1 = 'yes', 9 = 'unknown'
arrhythmia	History of Arrhythmia: 0 = 'no', 1 = 'yes', 9 = 'unknown'
PAD	History of PAD: 0 = 'no', 1 = 'yes', 9 = 'unknown'
hyperthyro	History of Hyperthyroidism: 0 = 'no', 1 = 'yes', 9 = 'unknown'
hyperUA	Hyperuricemia: 0 = 'no', 1 = 'yes', 9 = 'unknown'
blind	Blindness: 0 = 'no', 1 = 'yes', 9 = 'unknown'
cancer	Cancer: 0 = 'no', 1 = 'yes', 9 = 'unknown'
Cerebralhe	Intracerebral Hemorrhage: 0 = 'no', 1 = 'yes', 9 = 'unknown'
Cerebralin	Cerebral Infarction: 0 = 'no', 1 = 'yes', 9 = 'unknown'
TIA	History of TIA: 0 = 'no', 1 = 'yes', 9 = 'unknown'
MI	History of Myocardial Infarction: 0 = 'no', 1 = 'yes', 9 = 'unknown'
HF	History of Heart Failure: 0 = 'no', 1 = 'yes', 9 = 'unknown'
CAP	Carotid Artery Disease: 0 = 'no', 1 = 'yes', 9 = 'unknown'
hypothyroi	Hypothyroidism: 0 = 'no', 1 = 'yes', 9 = 'unknown'
RF	Renal Failure: 0 = 'no', 1 = 'yes', 9 = 'unknown'
Amputation	Amputation: 0 = 'no', 1 = 'yes', 9 = 'unknown'

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weightLoss	Weight Loss of More Than 5 Jin in the Past Year: 1 = 'no', 2 = 'yes', 9 = 'unknown'
fatherDM	Paternal Diabetes: 0 = 'no', 1 = 'yes', 9 = 'unknown'
motherDM	Maternal Diabetes: 0 = 'no', 1 = 'yes', 9 = 'unknown'
BSDM	Sibling Diabetes: 0 = 'no'
childDM	Offspring Diabetes: 0 = 'no'
DMfamilyhistory	Family History of Diabetes: 0 = 'no', 1 = 'yes'

Table 8. Diabetic Complications and Family History

Others. Table 9 illustrates 24 attributes, e.g., the Awareness of Diabetic Symptoms (known), Diet and Exercise for Diabetes Control (dietexcise), and complication information like the History of Diabetic Foot (DF) and Diabetic Retinopathy (DR), Tuning Fork Assessment of DPN (DPN by tuning fork), Monofilament Assessment of DPN (nlsdiag). Besides, screening conditions and classification indicators such as filter are set to precisely select research subjects.

Biochemical Parameter	**Description**
known	Awareness of Diabetic Symptoms: 1 = 'no', 2 = 'yes', 9 = 'unknown'
fatto	Obesity Predisposes to Diabetes: 1 = 'no', 2 = 'yes', 9 = 'unknown'
tochilren	Genetic Risk of Diabetes in Offspring: 1 = 'no', 2 = 'yes', 9 = 'unknown'
dietexcise	Diet and Exercise for Diabetes Control: 1 = 'no', 2 = 'yes', 9 = 'unknown'
compl	Diabetic Complications: 1 = 'no', 2 = 'yes', 9 = 'unknown'
check	Blood Glucose Testing History: 1 = 'no', 2 = 'yes', 9 = 'unknown'
glucoselev	Awareness of Blood Glucose: 1 = 'no', 2 = 'yes', 9 = 'unknown'
DF	History of Diabetic Foot: 2 = 'yes'
nlsdiag	Monofilament Assessment of DPN: 0 = 'no', 1 = 'yes'
DPN by tuning fork	Tuning Fork Assessment of DPN: 0 = 'no', 1 = 'yes'
DR	Diabetic Retinopathy: 1 = 'yes'
eyedia	Ophthalmology Diagnosis Summary
GAnum	Population size of the Jinuo ethnic group undergoing Glycated Albumin (GA) testing
filter	age >= 18 & age <= 75 & UMAUCR >= 0 & GFR5lev >= 0 & Fasting Plasma Glucose (FPG) >= 0 & HbA1c >= 0 & smoking >= 0 & drking >= 0 & Bpsys >= 0 & BMI > 1 & CDSdyelip >= 0 (FILTER), 0 = “Not Selected”, 1 = “Selected”
GAstudy	Diagnosis Based on Meeting All Three Criteria (No DM History)
A1cover6575	HbA1c is divided into three categories: 1 = '< 6. 5', 2 = '6. 5 to 7. 5', and 3 = '> 7. 5'
DMhighrisk	High-risk population for diabetes: 1 = 'yes'
DRyd	Fundus Photography with Possible DR: 1 = 'yes'
cataract	Cataract: 1 = 'yes'
anemia	Anemia: 0 = 'normal', 1 = 'anemia'
thalussemiaand	Thalassemia Initial Screening and: 0 = 'normal', 1 = 'suspected'

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thalussemiaor	Thalassemia initial screening or: 0 = 'normal', 1 = 'suspected'
thalussemiaHPLC	HPLC for thalassemia screening: 1 = 'suspected'

Table 9. Other relevant information

Technical Validation

To ensure the reliability, internal consistency, and overall quality of the dataset [12], a series of validations were conducted, focusing on statistical summaries, distributional plausibility and cross-variable coherence.

Statistical and Distributional Checks. Descriptive statistics were computed for all key variables (including BMI, FPG, PG2h, lipid profiles, renal function markers, insulin measures) to screen for data entry errors and assess basic data integrity. The unimodal distributions of continuous variables and the proportions of categorical variables were reviewed for consistency with general clinical expectations.

Validation of Inter-variable Relationships and Plausibility. To verify the biological plausibility and internal consistency among core measures, we examined the relationships between anthropometric and metabolic indicators.

In Figure 1, scatter plots between BMI and glycemic parameters were given. Linear regression analysis with 95% confidence intervals was performed to quantify their associations. The observed positive trends, along with the concentration of data points within established clinical thresholds for BMI and blood glucose, illustrated the physiological plausibility and internal coherence of these key variables.

Figure 1 goes here

Distribution plots of BMI and PG2h were created in Figure 2 to compare our cohort's characteristics with established epidemiological profiles for type 2 diabetes. The presence of a substantial proportion of subjects in higher BMI categories and the right-skewed distribution of PG2h, encompassing both controlled and elevated values, confirm that the dataset captures the heterogeneous and representative spectrum typical of real-world diabetic populations, thereby supporting its external validity.

Figure 2 goes here

In Figure 3, plots of paired FPG and PG2h values were inspected to assess the temporal and logical consistency within subjects. This check confirmed that the vast majority of paired measurements fall within clinically plausible and mutually consistent ranges, with no systematic measurement anomalies detected.

Figure 3 goes here

Validation of Cross-Variable Clinical Coherence. To assess the dataset's clinical face validity across different pathological dimensions, we examined the alignment of variables with expected clinical patterns.

A comparison of average glycemic levels across renal function stages was conducted in Figure 4. The observed pattern, where PG2h shows variation across stages while FPG remains relatively stable, falls within plausible bounds, supporting the dataset's internal clinical logic.

Figure 4 goes here

A bubble plot depicting the relationship between fasting insulin (FINS) and Homa-IR with linear regression was applied in Figure 5. The analysis revealed a strong positive correlation ($r = 0.970$, $p < 0.0001$), with FINS explaining 94% of the variance in Homa-IR ($R^2 = 0.940$). All data points fell within a physiologically plausible range, confirming the internal consistency and reliability of the insulin-related measures in the dataset.

Figure 5 goes here

In summary, these multi-faceted validation steps not only encompass distribution analysis, plausibility

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checks, internal consistency review and outlier screening, but also collectively affirm the dataset's robustness, consistency and suitability for further analysis.

Data Availability

This dataset is stored in the Zenodo repository with the URL: https://doi.org/10.5281/zenodo.18270337, and one can pick different categories discussed in the paper for various diabetes-related researches, e.g., diabetes pre-diagnosis, complication prediction or comparative analysis.

Code availability

The source code in this paper is published on Github: https://github.com/Zhoushanshen/Diabetes-dataset.

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Author contributions

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F.J. collected the raw dataset. J.L. provided the validation idea. H.Z. and Y.Z. conducted the experiment. J.L. and H.Z. wrote the manuscript, and prepared figures & tables. All authors analyzed the results. All authors reviewed and contributed to the final approval of the manuscript.

Competing interests

The authors declare no competing interests.

Figure legends

Fig. 1 Scatter plots of BMI versus blood glucose measurements with linear regression. (a) Relationship between BMI and fasting plasma glucose (FPG). Solid line represents the linear fit; shaded area indicates the 95% confidence interval. (b) Relationship between BMI and 2-hour postprandial glucose (PG2h). Regression line and 95% confidence interval are shown. Clinical thresholds for BMI categories and glycemic status are indicated by vertical dashed lines and horizontal-colored bands, respectively.

Fig. 2 Distributions of clinical measurements in the patient cohort. (a) Distribution of Body Mass Index (BMI). (b) Distribution of 2-hour Postprandial Glucose (PG2h).

Fig. 3 Longitudinal blood glucose measurements in diabetic patients. The line graph displays sequential measurements of Fasting Plasma Glucose (FPG, blue dots) and 2-hour Postprandial Glucose (PG2h, orange squares) across sample order. FPG values are relatively stable within the 0-10 mmol/L range, while PG2h values show greater dispersion and generally fall within clinically relevant ranges.

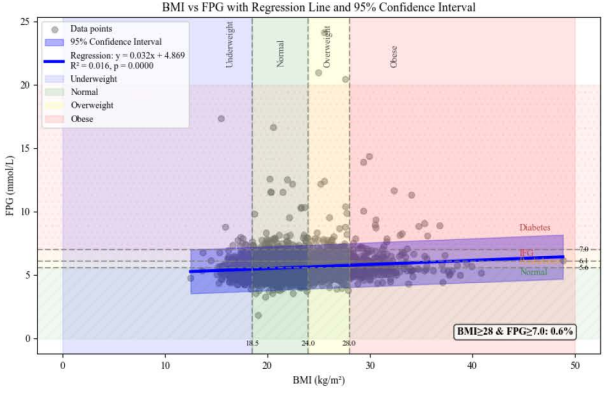
Fig. 4 Stacked bar chart of average FPG (blue) and PG2h (orange) across five renal function stages (GFR5lev 0-4).

Fig. 5 Bubble plot of FINS versus Homa-IR with linear regression. The size of each bubble is proportional to the product of FINS and Homa-IR values. The solid red line indicates the linear fit, and the shaded band represents the 95% confidence interval. The Pearson correlation coefficient (r), coefficient of determination (R^2), and p-value are displayed on the graph.

ARTICLE IN PRESS

ARTICLE IN PRESS

a



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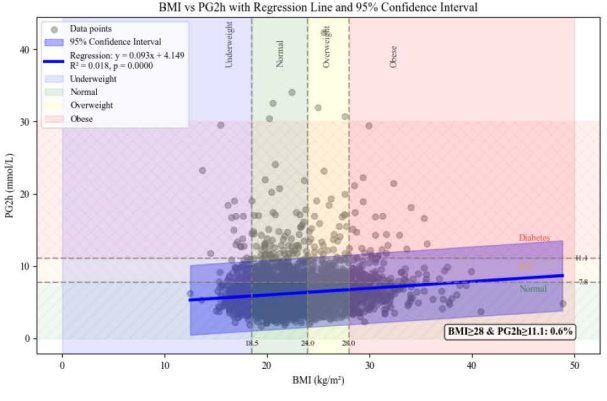

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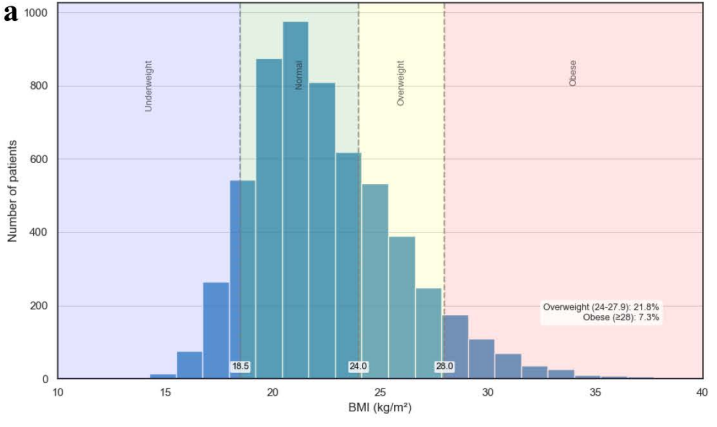
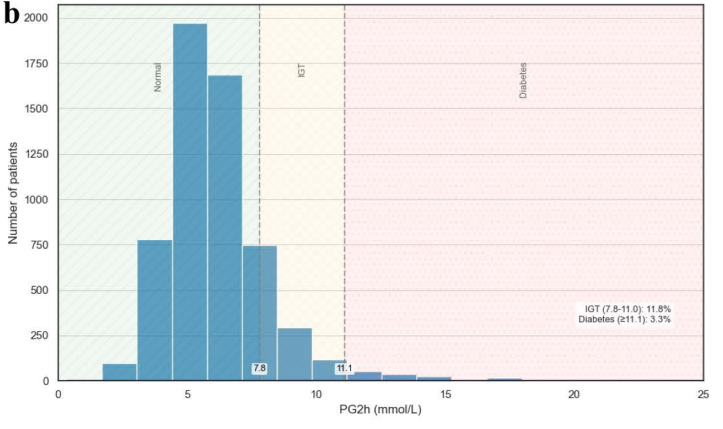
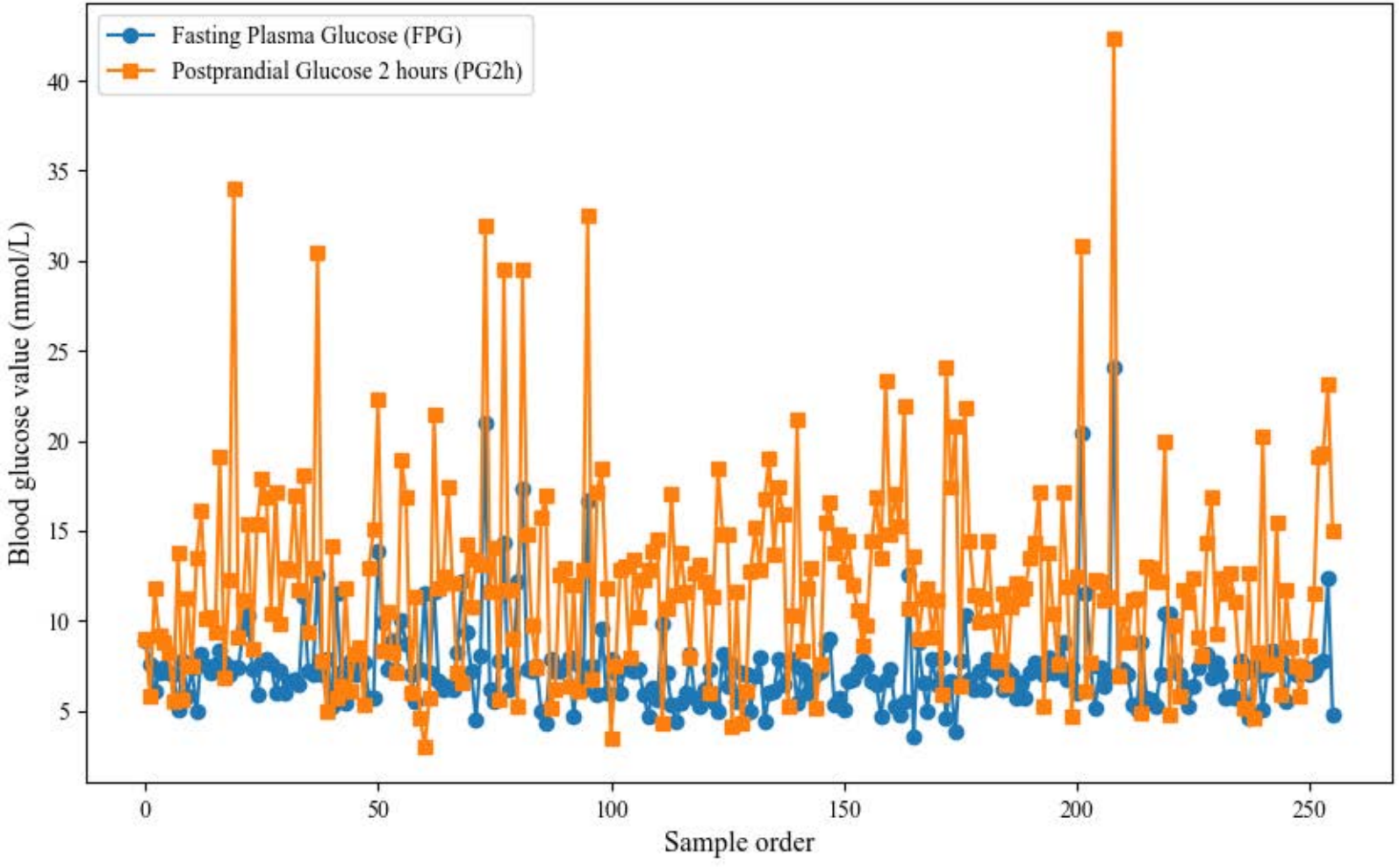
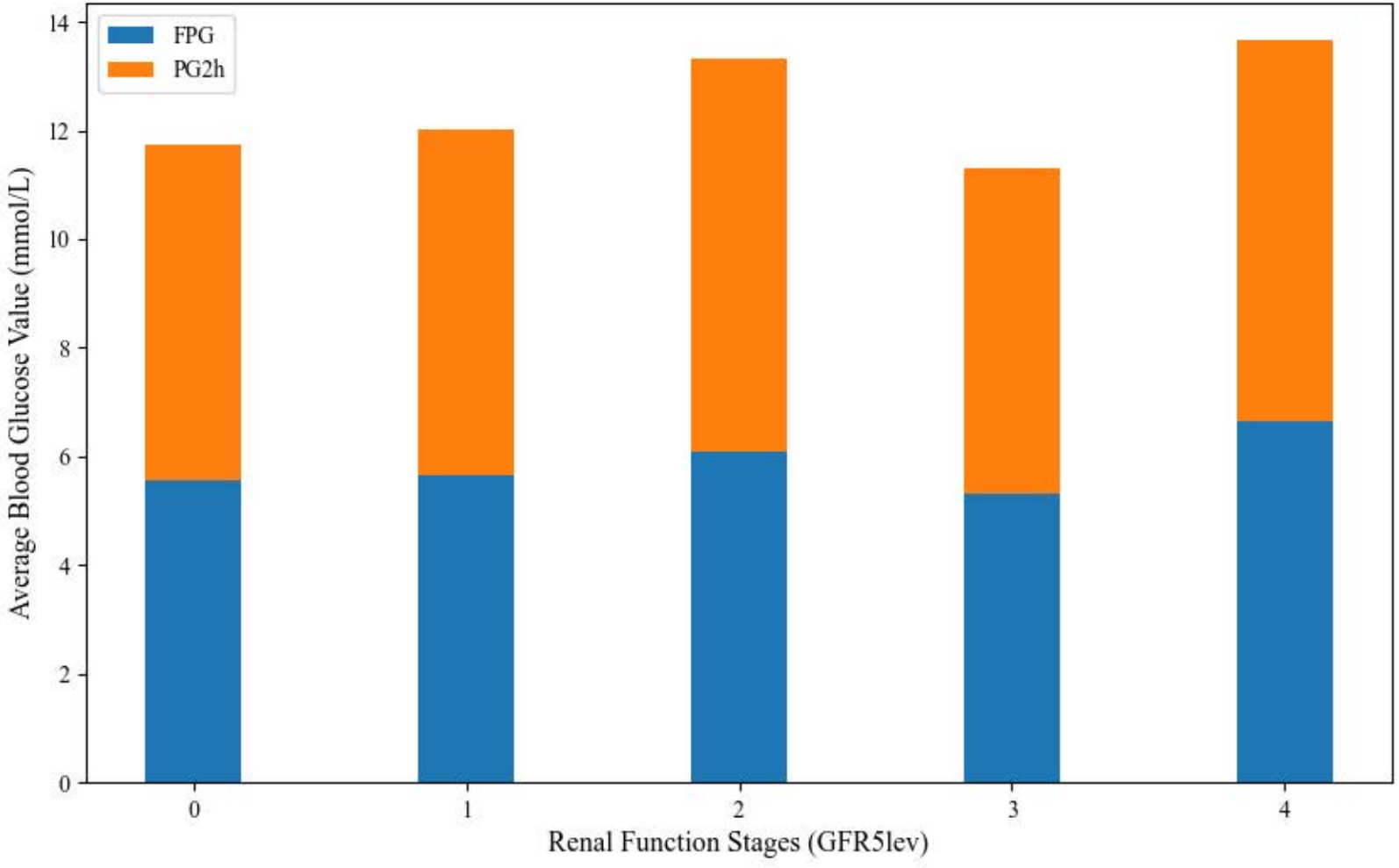
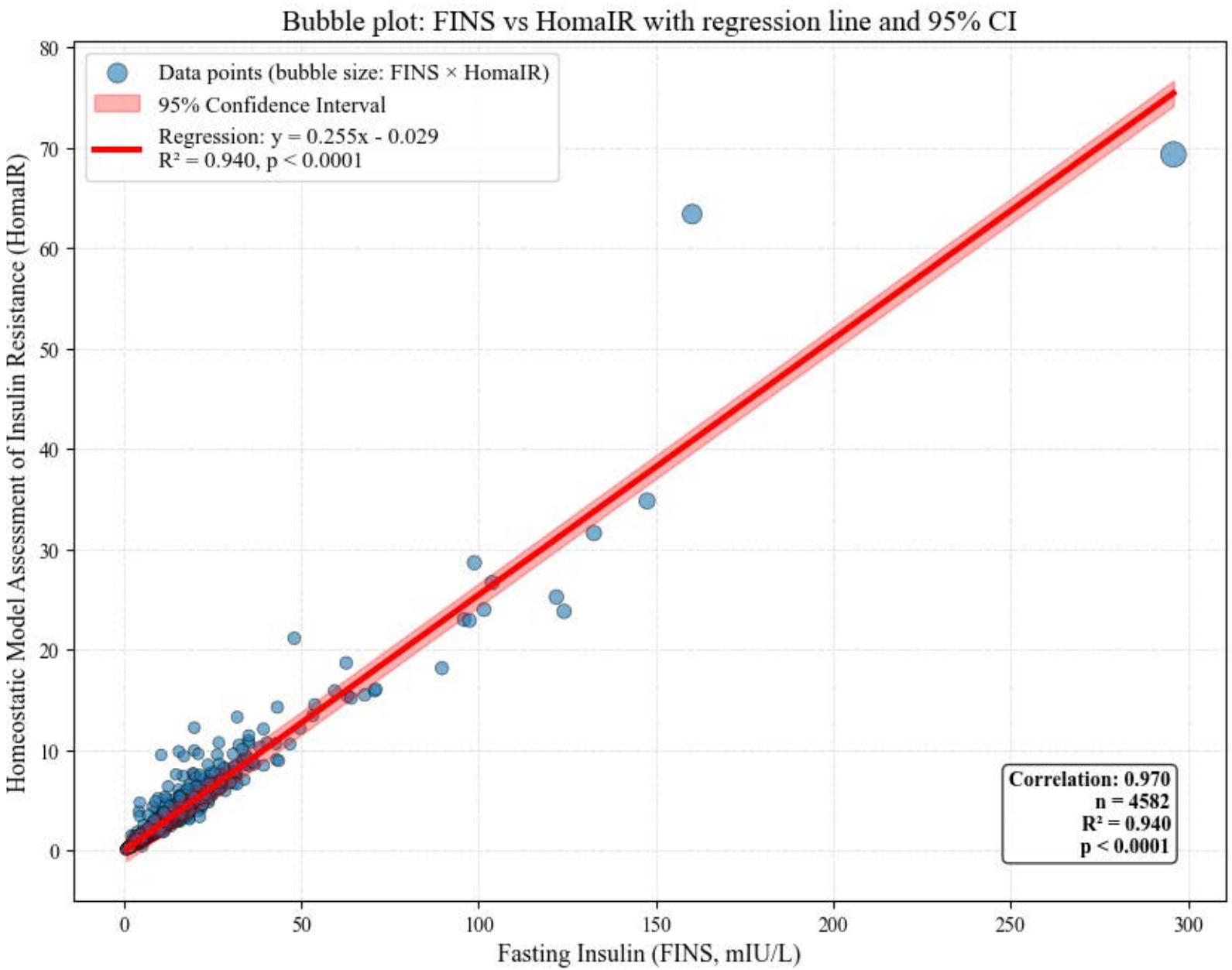





Fasting Plasma Glucose (FPG)
Postprandial Glucose 2 hours (PG2h)
Blood glucose value (mmol/L)
Sample order
0
50
100
150
200
250
5
10
15
20
25
30
35
40


FPG
PG2h
Average Blood Glucose Value (mmol/L)
0
2
4
6
8
10
12
14
0
1
2
3
4
Renal Function Stages (GFR5lev)


Bubble plot: FINS vs HomaIR with regression line and 95% CI
Data points (bubble size: FINS × HomaIR)
95% Confidence Interval
Regression: y = 0.255x - 0.029
R² = 0.940, p < 0.0001
Correlation: 0.970
n = 4582
R² = 0.940
p < 0.0001
Homeostatic Model Assessment of Insulin Resistance (HomaIR)
Fasting Insulin (FINS, mIU/L)
0
10
20
30
40
50
60
70
80
0
50
100
150
200
250
300